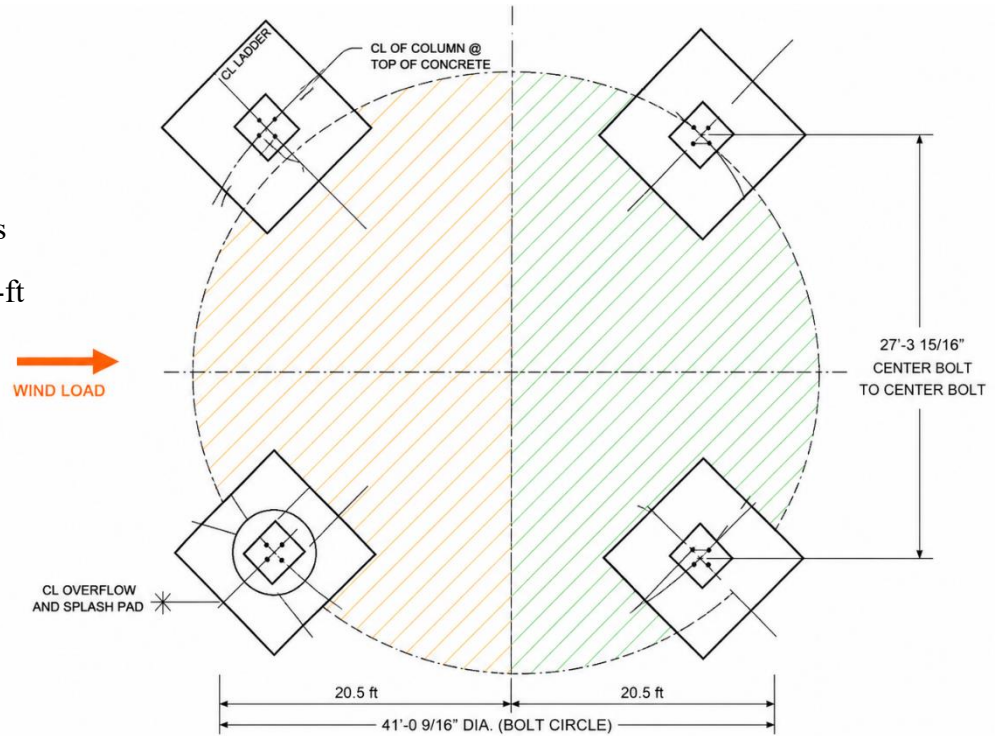


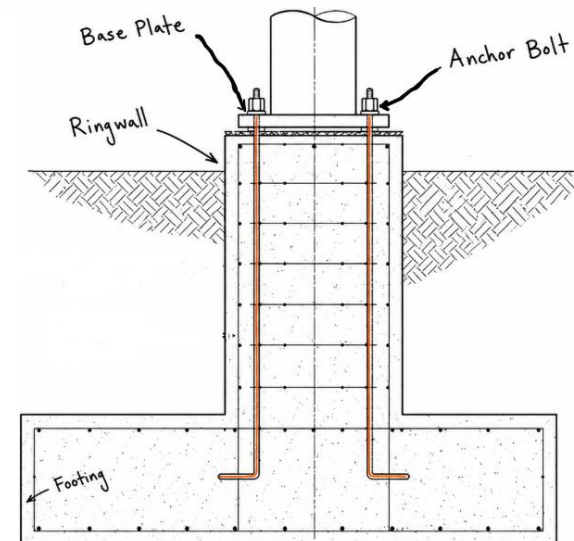
1. Anchor Bolts segment input

Input	Symbol	Value
Factored axial load	P_u	1609.756 kips
Factored moment	M_u	5673.437 kip-ft
Factored shear	V_u	52.30 kips
Number of anchor bolts	N_b	4
Cone Diameter	D_{cone}	41'-0.5625"
Anchor bolt diameter	d_b	1.75 in
Effective embedment depth	h_{ef}	40 in
Concrete strength	f_c	4000 psi
Anchor bolt yield strength	F_y	36 ksi
Anchor bolt ultimate strength	F_u	58 ksi
Edge distance	$e_x = e_y$	3 in
Pedestal Size	$(B \times L)$	39 in $\times$ 39 in
Bolt arrangement	$(b \times l)$	12 in $\times$ 12 in
Strength reduction factors	ϕ_t, ϕ_v	0.75, 0.75



COMPRESSION AND TENSION ZONES – PLAN VIEW
(CIRCLE AT CENTER OF EACH FOUNDATION/COLUMN)

LEG KEY
1 – Leg 1 (Top)
2 – Leg 2 (Right)
3 – Leg 3 (Bottom)
4 – Leg 4 (Left)



STEP 1 — DETERMINE TENSION SIDE LEGS

For overturning, assume:

- 2 legs in compression
- 2 legs in tension

Thus.....nt=2

STEP 2 — DETERMINE LEG RADIUS

$$D = 41' - 0\frac{9}{16}"$$

Radius:

$$r = 20.523'$$

Convert:

$$r = 246.28"$$

STEP 3 — TENSION FORCE FROM OVERTURNING

$$T = \frac{M_u}{r}$$

Convert moment:

$$M_u = 5673.44 \times 12$$

$$M_u = 68081.3 \text{ kip-in}$$

Now:

$$T = \frac{68081.3}{246.28}$$

$$T = 276.4 \text{ kips}$$

This is total overturning uplift.

Shared by 2 tension legs:

$$T_{leg} = \frac{276.4}{2}$$

$$T_{leg} = 138.2 \text{ kips}$$

$$T_{bolt} = \frac{138.2}{2}$$

$$T_{bolt} = 69.1 \text{ kips}$$

STEP 4 — TENSION PER BOLT

STEP 5 — REQUIRED STEEL AREA

LRFD:

$$\phi = 0.75$$

Required steel area:

$$A_s = \frac{T_u}{\phi F_y}$$

$$A_s = \frac{69.1}{0.75(36)}$$

$$A_s = 2.56 \text{ in}^2$$

STEP 6 — SELECT BOLT DIAMETER

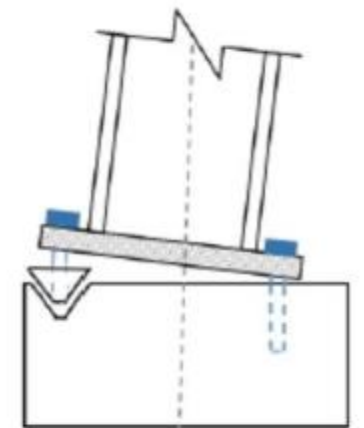
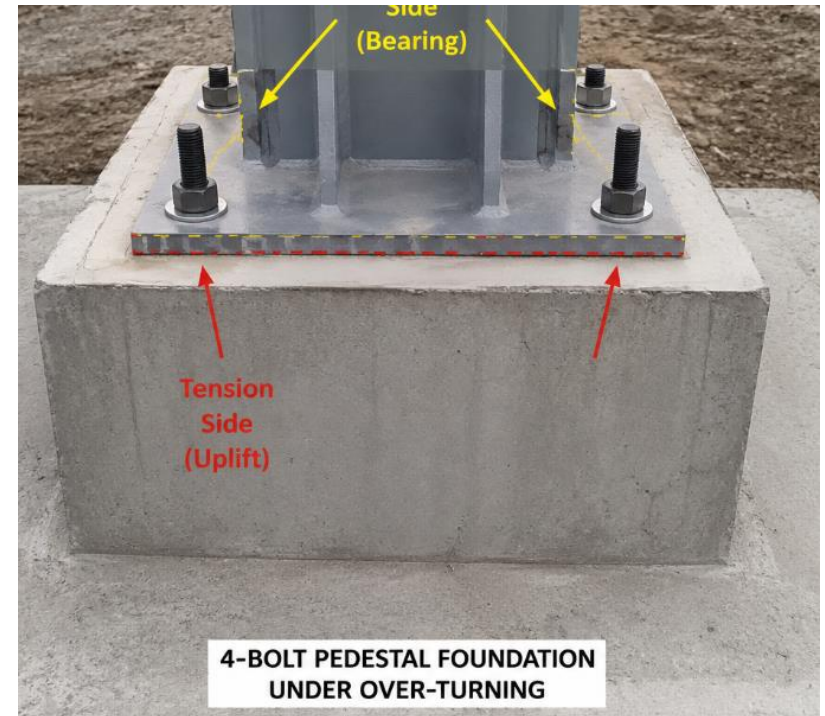
Diameter	Area
1.75 in	1.90 in ²
2.0 in	2.50 in ²
2.25 in	3.25 in ²

Now:

$$A_s = 2.56 \text{ in}^2$$

Thus:

- 2 in rod is NO LONGER sufficient



Anchor bolt failure

- 2.25 in rod is required

SELECT

2.25" diameter anchor rod

Area:

$$A_b = 3.25 \text{ in}^2$$

STEP 7 — CHECK STEEL TENSION CAPACITY

$$\phi N_s = \phi A_b F_y$$

$$\phi N_s = 0.75(3.25)(36)$$

$$\phi N_s = 87.75 \text{ kips}$$

Check:

$$87.75 > 69.1 \dots \dots \dots \text{PASS}$$

STEP 8 — CHECK STEEL SHEAR CAPACITY

Shear per pedestal:

$$V_{leg} = \frac{52.3}{4}$$

$$V_{leg} = 13.08 \text{ kips}$$

Shear per bolt:

$$V_{bolt} = \frac{13.08}{4}$$

$$V_{bolt} = 3.27 \text{ kips}$$

Steel shear capacity:

$$\phi V_s = 0.75(0.6)(36)(3.25)$$

$$\phi V_s = 52.65 \text{ kips}$$

Check:

$$52.65 > 3.27 \dots\dots\dots \text{PASS}$$

STEP 9 — CONCRETE BREAKOUT CHECK

ACI approximate breakout:

$$N_{cb} = \phi k_c \sqrt{f'_c} h_{ef}^{1.5}$$

Use:

- $k_c = 24$
- $\phi = 0.75$

Compute:

$$\sqrt{4000} = 63.25$$

$$40^{1.5} = 253$$

$$N_{cb} = 0.75(24)(63.25)(253)$$

$$N_{cb} = 288000 \text{ lb}$$

$$N_{cb} = 288 \text{ kips}$$

Check:

$$288 > 69.1 \dots\dots\dots \text{PASS}$$

STEP 10 — PULLOUT CHECK

For 2.25 in rod:

Assume:

- 5 in square plate washer

$$A_{brg} = 25 \text{ in}^2$$

Pullout:

$$N_p = 8A_{brg}f'_c$$

$$N_p = 8(25)(4000)$$

$$N_p = 800000 \text{ lb}$$

$$N_p = 800 \text{ kips}$$

Check:

$$800 > 69.1 \dots \dots \dots \text{PASS}$$

STEP 11 — PRYOUT CHECK

ACI:

$$V_{cp} = 1.0N_{cb}$$

$$V_{cp} = 288 \text{ kips}$$

Check:

$$288 > 3.27 \dots \dots \dots \text{PASS}$$

STEP 12 — INTERACTION CHECK

$$\left(\frac{N_u}{\phi N_n} \right)^2 + \left(\frac{V_u}{\phi V_n} \right)^2 \leq 1.0$$

Compute:

$$\begin{aligned} & \left(\frac{69.1}{87.75} \right)^2 + \left(\frac{3.27}{52.65} \right)^2 \\ & = 0.620 + 0.0039 \end{aligned}$$

$$= 0.624$$

Check:

$$0.624 < 1.0 \dots \dots \dots \text{PASS}$$

STEP 13 — EDGE DISTANCE CHECK

Pedestal:

$$39"$$

Bolt spacing:

$$12"$$

Edge distance:

$$\frac{39 - 12}{2}$$

$$= 13.5"$$

Required minimum:

$$7d = 7(2.25)$$

$$= 15.75"$$

Current pedestal is deficient.

RECOMMENDATION

Increase pedestal size to:

$$44" \times 44"$$